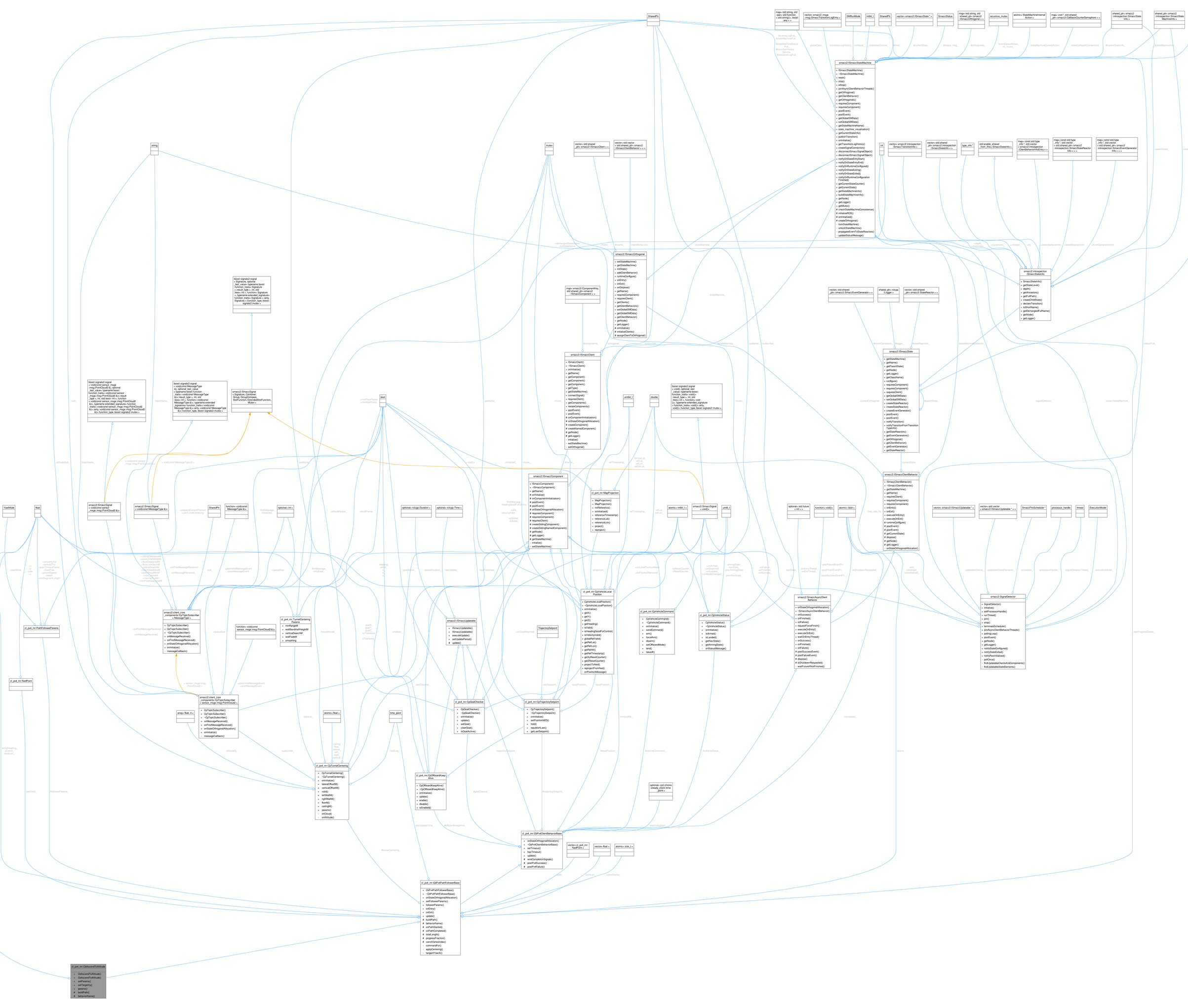

SharedPtr
map< std::string, std::pair< std::function< std::string()>, bool > >
vector< smacc2::msgs::SmTransitionLogEntry >
SMRunMode
int64_t
SharedPtr
vector< smacc2::ISmaccState * >
SmaccStatus
map< std::string, std::shared_ptr< smacc2::ISmaccOrthogonal > >
recursive_mutex
atomic< StateMachineInternalAction >
map< void *, std::shared_ptr< smacc2::CallbackCounterSemaphore > >
shared_ptr< smacc2::introspection::SmaccStateInfo >
shared_ptr< smacc2::introspection::SmaccMachineInfo >
SharedPtr
smacc2::ISmaccStateMachine
+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ reset()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponents()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ getTransitionLogHistory()
+ createSignalConnections()
+ disconnectSmaccSignalObject()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfigurationFinished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
checkStateMachineConsistence()
initializePOS()
oninitialized()
createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStateMessage()
string
mutex
vector< std::shared_ptr< smacc2::ISmaccClient > >
vector< std::shared_ptr< smacc2::ISmaccClientBehavior > >
int
vector< smacc2::introspection::SmaccTransitionInfo >
vector< std::shared_ptr< smacc2::introspection::SmaccStateInfo > >
type_info *
std::enable_shared_from_this< SmaccStateInfo >
map< const std::type_info *, std::vector< smacc2::introspection::ClientBehaviorInfoEntry > >
map< const std::type_info *, std::vector< std::shared_ptr< smacc2::introspection::SmaccReactorInfo > > >
map< const std::type_info *, std::vector< std::shared_ptr< smacc2::introspection::SmaccEventGeneratorInfo > > >
smacc2::ISmaccOrthogonal
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
oninitialize()
initializeClients()
assignClientToOrthogonal()
smacc2::introspection::SmaccStateInfo
+ SmaccStateInfo()
+ getStateLevel()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()
map< smacc2::ComponentKey, std::shared_ptr< smacc2::ISmaccComponent > >
vector< std::shared_ptr< smacc2::ISmaccEventGenerator > >
shared_ptr< rclcpp::Logger >
vector< std::shared_ptr< smacc2::IStateReactor > >
smacc2::ISmaccState
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransitionObject()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()
smacc2::ISmaccClient
+ ISmaccClient()
+ ~ISmaccClient()
+ oninitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
onComponentInitialization()
onStateOrthogonalAllocation()
createComponent()
createNamedComponent()
getNode()
getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()
boost::signals2::signal< void(), optional_last_value< boost::function_traits< void() >::result_type >, int, std::less< int >, boost::function_traits< void() >::arity, boost::signals2::mutex >
boost::signals2::signal< void(const MessageType &), optional_last_value< typename boost::function_traits< void(const MessageType &) >::result_type >, int, std::less< int >, boost::function< void(const MessageType &) >, typename extended_signature< boost::function_traits< void(const MessageType &) >::arity, void(const MessageType &) >::function_type, boost::signals2::mutex >
smacc2::SmaccSignal< Signature, Combiner, Group, GroupCompare, SlotFunction(), ExtendedSlotFunction, Mutex >
boost::signals2::signal< void(const sensor_msgs::msg::PointCloud2 &), optional_last_value< typename boost::function_traits< void(const sensor_msgs::msg::PointCloud2 &) >::result_type >, int, std::less< int >, boost::function< void(const sensor_msgs::msg::PointCloud2 &) >, typename extended_signature< boost::function_traits< void(const sensor_msgs::msg::PointCloud2 &) >::arity, void(const sensor_msgs::msg::PointCloud2 &) >::function_type, boost::signals2::mutex >
smacc2::SmaccSignal< void(const sensor_msgs::msg::PointCloud2 &) >
smacc2::SmaccSignal< void(const MessageType &) >
SharedPtr
function< void(const MessageType &) >
optional< int >
bool
uint64_t
double
boost::signals2::signal< void(), optional_last_value< typename boost::function_traits< void() >::result_type >, int, std::less< int >, boost::function< void() >, typename extended_signature< boost::function_traits< void() >::arity, void() >::function_type, boost::signals2::mutex >
smacc2::ISmaccComponent
+ ISmaccComponent()
+ ~ISmaccComponent()
+ getName()
+ oninitialize()
onComponentInitialization()
postEvent()
postEvent()
onStateOrthogonalAllocation()
requiresComponent()
requiresComponent()
requiresClient()
createSiblingComponent()
createSiblingNamedComponent()
getNode()
getLogger()
getStateMachine()
- initialize()
- setStateMachine()
cl_px4::mr::MapProjection
+ MapProjection()
+ MapProjection()
+ initReference()
+ isInitialized()
+ referenceTimestamp()
+ referenceLat()
+ referenceLon()
+ project()
+ reproject()
YawMode
float
smacc2::SmaccSignal< void() >
uint8_t
optional< std::future< int > >
function< void() >
atomic< bool >
smacc2::ISmaccClientBehavior
+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
+ runtimeConfigure()
postEvent()
postEvent()
getCurrentState()
dispose()
getNode()
getLogger()
- onStateOrthogonalAllocation()
vector< smacc2::ISmaccUpdatable * >
vector< std::vector< smacc2::ISmaccUpdatable * > >
SmaccRosScheduler *
processor_handle
thread
ExecutionModel
cl_px4::mr::CpVehicleLocalPosition
+ CpVehicleLocalPosition()
+ ~CpVehicleLocalPosition()
+ onInitialize()
+ getX()
+ getY()
+ getZ()
+ getHeading()
+ isValid()
+ isHeadingGoodForControl()
+ isVelocityValid()
+ globalRefValid()
+ getRefLat()
+ getRefLon()
+ getRefAlt()
+ getRefTimestamp()
+ getXyResetCounter()
+ getZResetCounter()
+ projectToNed()
+ reprojectFromNed()
- onPositionMessage()
cl_px4::mr::CpVehicleCommand
+ CpVehicleCommand()
+ ~CpVehicleCommand()
+ oninitialize()
+ sendCommand()
+ arm()
+ forceArm()
+ disarm()
+ setOffboardMode()
+ land()
+ takeoff()
cl_px4::mr::CpVehicleStatus
+ CpVehicleStatus()
+ ~CpVehicleStatus()
+ oninitialize()
+ isArmed()
+ isLanded()
+ getNavState()
+ getArmingState()
- onStatusMessage()
smacc2::SmaccAsyncClientBehavior
+ onStateOrthogonalAllocation()
+ ~SmaccAsyncClientBehavior()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
postSuccessEvent()
postFailureEvent()
dispose()
isShutdownRequested()
+ waitFutureIfNotFinished()
smacc2::SignalDetector
+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ joinAsyncClientBehaviorThreads()
+ pollingLoop()
+ postEvent()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyRosInitialized()
- pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()
cl_px4::mr::PathFollowerParams
cl_px4::mr::NedPoint
cl_px4::mr::FlightPatternAscendParams
cl_px4::mr::ClAscendToAltitude
+ ClAscendToAltitude()
+ ClAscendToAltitude()
+ setParams()
+ setTarget(xy)
+ params()
buildPath()
behaviorName()
smacc2::client_core_components::CpTopicSubscriber< MessageType >
+ CpTopicSubscriber()
+ CpTopicSubscriber()
+ ~CpTopicSubscriber()
+ onMessageReceived()
+ onFirstMessageReceived()
+ onStateOrthogonalAllocation()
+ oninitialize()
- messageCallback()
function< void(const sensor_msgs::msg::PointCloud2 &)>
cl_px4::mr::TunnelCentering Params
+ minRangeM
+ waitBandHalfHeightM
+ verticalSlackM
+ waitFadeM
+ smoothing
smacc2::client_core_components::CpTopicSubscriber< sensor_msgs::msg::PointCloud2 >
+ CpTopicSubscriber()
+ CpTopicSubscriber()
+ ~CpTopicSubscriber()
+ onMessageReceived()
+ onFirstMessageReceived()
+ onStateOrthogonalAllocation()
+ oninitialize()
- messageCallback()
array< float, 4 >
atomic< float >
time_point
cl_px4::mr::CpTunnelCentering
+ CpTunnelCentering()
+ ~CpTunnelCentering()
+ oninitialize()
+ lateralOffsetM()
+ verticalOffsetM()
+ valid()
+ leftWallM()
+ rightWallM()
+ floorM()
+ ceilingM()
+ params()
+ onCloud()
+ onAttitude()
cl_px4::mr::CpOffboardKeep Alive
+ CpOffboardKeepAlive()
+ ~CpOffboardKeepAlive()
+ oninitialize()
+ update()
+ enable()
+ disable()
+ isEnabled()
cl_px4::mr::CpGoalChecker
+ CpGoalChecker()
+ ~CpGoalChecker()
+ oninitialize()
+ update()
+ setGoal()
+ clearGoal()
+ isGoalActive()
cl_px4::mr::CpTrajectorySetpoint
+ CpTrajectorySetpoint()
+ ~CpTrajectorySetpoint()
+ oninitialize()
+ setPositionNED()
+ hold()
+ republishLast()
- getLastSetpoint()
smacc2::ISmaccUpdatable
+ ISmaccUpdatable()
+ ISmaccUpdatable()
+ executeUpdate()
setUpdatePeriod()
update()
TrajectorySetpoint
cl_px4::mr::ClPx4PathFollowerBase
+ ClPx4PathFollowerBase()
+ ~ClPx4PathFollowerBase()
+ onStateOrthogonalAllocation()
+ setFollowerParams()
+ followerParams()
+ onEntry()
+ onExit()
+ update()
buildPath()
behaviorName()
onPathStarted()
onPathCompleted()
totalLength()
progressFraction()
carrotVertexIndex()
- commandFor()
- applyCornering()
- tangentTaperAt()
cl_px4::mr::ClPx4ClientBehaviorBase
+ onStateOrthogonalAllocation()
+ ~ClPx4ClientBehaviorBase()
+ setTimeout()
+ hasTimeout()
+ update()
wireCompletionSignals()
postPx4Success()
postPx4Failure()
vector< cl_px4::mr::NedPoint >
vector< float >
atomic< size_t >
optional< std::chrono::steady_clock::time_point >
cl_px4::mr::ClAscendToAltitude
+ ClAscendToAltitude()
+ ClAscendToAltitude()
+ setParams()
+ setTarget(xy)
+ params()
buildPath()
behaviorName()